NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Blockchain Design Architecture (CS740), MID-SEM EXAM, 2nd Semester-M.Tech

Note:

I. Answer all questions.
II. Write all parts of one question in one place.

Total: *20 marks*
Duration: *1:30 Hr.*
Date: *February 25, 2026*

Q No.	Question	Part marks
1	a. How do smart contracts work? b. In bitcoin generation, after how many blocks is the block creation rate adjusted? Why? c. What is Double Spending Problem? Explain some more attacks on PoW based system and their solutions.	2+1+3=**6**
2	a. What are the correctness measures of *Distributed Consensus Protocol*? b. What is *Mining Difficulty*? State its relation with *Hashrate.* c. What do you understand by *Bitcoin Anonymity*?	1+3+2=**6**
3	a. Explain the *Monopoly Problem* in Proof-of-Work (PoW) based blockchain systems. Discuss how computational resources influence mining power and how *monopoly* can increase over time. What are the possible consequences for miners, users, and the overall network? b. A message **M** with a length of exactly 1,450 bits is prepared for **SHA-256** hashing. Calculate the specific number of **padding zeros (k)** that must be appended to ensure the message meets the block size requirements of the NIST FIPS 180-4 standard. After preprocessing (padding and length encoding), determine: i. The total number of 512-bit message blocks (N) processed by the compression function.	2+2=**4**

	ii. The total number of initial 32-bit message words obtained by partitioning all padded message blocks (16 words per block).	
4	A system records 10 unique transactions ($T_1 \ldots T_{10}$). The internal hierarchy and pairing logic of the resulting Merkle Tree are defined by the following Post-Order Traversal: H1, H2, H12, H3, H4, H34, H1234, H5, H6, H56, H8, H7, H87, H5687, H12345687, H9, H10, H910, ROOT a. Draw the complete Merkle Tree diagram based on the logic provided in the traversal string. Label every leaf node and every intermediate concatenation node. b. Calculate the total number of hashing operations required to generate the ROOT from the 10 initial transactions. c. Identify which specific hash was "promoted" to a higher level without a sibling pair, and specify the level at which this occurred to eventually reach the Root node. d. A light client needs to verify the inclusion of transaction T_7. List the specific Audit Path (the sibling hashes) that the full node must provide, and indicate their orientation (Left or Right) relative to the client's calculated hashes. **<u>Note:</u>** • Assume a binary Merkle Tree where child hashes are combined strictly in left-to-right order. When an odd number of nodes exists at any level, the last node is promoted unchanged to the next level (not duplicated). • Every parent node : Parent = Hash(Left_child || Right_child) (Left concatenated before Right) • Each transaction T_i is hashed individually to produce leaf hash H_i. • The tree must be reconstructed exactly from the provided postorder traversal under the above pairing rules. • Level 0 = leaf level. Level number increases upward toward the root. • While counting Hash operation you need to include Leaf hashing .	1+1+1+1=**4**

-: Good Luck :-